

JUNYEOP BAEK

Ph.D. Student @ KAIST

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RESEARCH INTEREST

My research goal is to build “An embodied machine that can develop intelligence throughout its lifetime, just like humans.” Specifically, the machine should establish and grow its own lens of worlds, goals, and values in open-ended environments — with interpretability and alignment as guiding concerns.

My current path toward this vision is to build **compositional world models** — understanding the world as reusable structure — and the **recursive reasoning** that runs over them to generalize beyond experience.

I pursue the following research questions, with **Q1** as my current focus throughout the PhD:

- **Q1. How can AI understand the world more systematically and compositionally? – Systematic World Understanding & Generalization (current focus)**
- *Q2. How can AI continuously accumulate and apply knowledge/skills? – Continual Learning*
- *Q3. Can AI develop its own goals following intrinsic motivation? – Autotelic Agents*
- *Q4. How can AI maintain autonomy while sharing human values? – Human-Value Alignment*

EDUCATION

Korea Advanced Institute of Science and Technology(KAIST)

Mar. 2025 – Present

Ph.D. in Computer Science - **Advisor:** Sungjin Ahn

Daejeon, Korea

Korea Advanced Institute of Science and Technology(KAIST)

Mar. 2023 – Feb. 2025

M.S. in Artificial Intelligence - **Advisor:** Sungjin Ahn

Daejeon, Korea

- **Thesis:** Dreamweaver: Learning Compositional World Models from Pixels

Hanyang University(ERICA)

Mar. 2014 – Feb. 2018

B.S. in Robot Engineering - **GPA** - 4.29 / 4.5

Ansan, Korea

- *Summa Cum Laude*
- *Capstone Project1: Disaster and Rescue Robot Simulation Modeling(Darrsm) - 2nd Prize in College*
- *Capstone Project2: Walk Yourself, Toddler!(WalkYTo) - 3rd Prize in Department*

RESEARCH EXPERIENCE

Visiting Researcher

Jun. 2025 – Aug. 2025

Mila-Quebec AI Institute

Montreal, Canada

- Collaborating with *Yoshua Bengio*’s research group and *LawZero* team members
- Conducting research project on **Mechanistic Interpretability and Chain-of-Thought Faithfulness in LLMs**
- **Main Collaborators:** *Joumana Ghosn, Pierre-Luc St-Charles, Marc-Antoine Rondeau, Minsu Kim*

HONORS & AWARDS

- **Best Paper Award**, ICML 2026 Workshop on Compositional Learning (2026)
- **Oral Presentation (Top 0.7%, 168/23918)**, ICML 2026
- **Silver Reviewer**, ICML 2026
- **Outstanding Teaching Assistant Award**, KAIST (2025)

PUBLICATIONS

1. Doojin Baek*, Gyubin Lee*, **Junyeob Baek**, Hosung Lee, and Sungjin Ahn. "Learning to Theorize the World from Observation". In: *International Conference on Machine Learning (ICML)* (2026). **Oral Presentation (Top 0.7%) • Best Paper Award**, ICML 2026 Workshop on Compositional Learning. (*Equal contribution)
2. **Junyeob Baek***, Mingyu Jo*, Minsu Kim, Mengye Ren, Yoshua Bengio, and Sungjin Ahn. "Generative Recursive Reasoning". *Preprint* (2026). (*Equal contribution)
3. **Junyeob Baek**, Hosung Lee, Christopher Hoang, Mengye Ren, and Sungjin Ahn. "Discrete JEPA: Learning Discrete Token Representations without Reconstruction". In: *International Conference on Machine Learning (ICML) Tokenization Workshop* (2025).
4. Donghwan Chi, Hyomin Kim, Yoojin Oh, Yongjin Kim, Donghoon Lee, Daejin Jo, Jongmin Kim, **Junyeob Baek**, Sungjin Ahn, and Sungwoong Kim. "Slot-MLLM: Object-Centric Visual Tokenization for Multimodal LLM". *Preprint, Under Review* (2025).
5. **Junyeob Baek**, Yi-Fu Wu, Gautam Singh, and Sungjin Ahn. "Dreamweaver: Learning Compositional World Models from Pixels". In: *International Conference on Learning Representation (ICLR)* (2025).
6. Chang Chen, **Junyeob Baek**, Fei Deng, Kenji Kawaguchi, Caglar Gulcehre, and Sungjin Ahn. "PlanDQ: Hierarchical Plan Orchestration via D-Conductor and Q-Performer". In: *International Conference on Machine Learning (ICML)* (2024).

THESIS

- **Junyeob Baek**. "Dreamweaver: Learning Compositional World Models from Pixels". Master's Thesis, KAIST, Advisor: Sungjin Ahn (Feb. 2025).

ACADEMIC SERVICE

Conference/Journal Reviewer

- International Conference on Learning Representations (ICLR) 2025–2026
- International Conference on Machine Learning (ICML) 2026 — **Silver Reviewer**
- Conference on Neural Information Processing Systems (NeurIPS) 2026
- Transactions on Machine Learning Research (TMLR) 2026–

Workshop Organizer

- 2nd KAIST-Mila Prefrontal AI Workshop (6 invited speakers incl. Yoshua Bengio), [Link](#)

TEACHING EXPERIENCE

Teaching Assistant

CS30707: Introduction to Reinforcement Learning, KAIST

Spring 2026

Teaching Assistant

CS672: Reinforcement Learning, KAIST

Fall 2025

Teaching Assistant

CS377: Introduction to Reinforcement Learning, KAIST

Spring 2025

- Outstanding Teaching Assistant Award recipient

PROFESSIONAL EXPERIENCE

MakinaRocks

Jun. 2021 – Jun. 2022

ML Research Engineer(Robotics Focused)

Seoul, Korea

- **Optimization of Welding Task and Motion Planning in Vehicle Manufacturing Process**
 - * Challenges: 1. Large-scale and high complexity in Optimization Problem. 2. Required high-accuracy planning in highly cluttered environments. 3. Required tight Time Window for control
 - * [*Contribution 1*] GA-based Multi-objective Optimization Module for Welding-point Assignment
 - * [*Contribution 2*] Multi-Robot Interlock Scheduling System with Swept-volume-based collision detection

Syscon, Co., Ltd.

Dec. 2017 – Jun. 2021

Robotics Software Engineer/Research Strategy Manager

Incheon, Korea

- **AMR Navigation System Design & Product Development**
- [*Field Exp.*] **Successful Deployment for Smart Factory Project**
 - * More than 300 delivery tasks per day with 14 robots
- [*Research Exp.*] **Learning-based High-accuracy Object Pose Estimator(Dock Pose Estimation)**
- [*Research Exp.*] **Distilled Meta Policy Navigation System for Multi-behavior Skillset and its Strategies**